

Exploratory Data Analysis (EDA) Report

Delinquency Prediction Dataset

This report summarizes the exploratory data analysis performed on the Delinquency Prediction dataset to assess data quality, identify anomalies, handle missing values, and detect key risk indicators influencing customer delinquency.

1. Dataset Overview

- Dataset contains 500 customer records and 19 variables.
- Target variable: Delinquent_Account (0 = Non-Delinquent, 1 = Delinquent).
- The dataset contains both numerical and categorical customer attributes.

2. Data Quality Assessment

- Missing values were identified in Income (39), Loan_Balance (29), and Credit_Score (2).
- No duplicate records were detected.
- Inconsistent employment labels (EMP, Employed, employed) were found and should be standardized.
- No significant outliers were detected in the numerical variables.

3. Missing Value Treatment

- Income: Missing values were imputed using the median.
- Loan_Balance: Median imputation was applied.
- Credit_Score: Median imputation was used due to the small number of missing values.

4. Key Patterns and Anomalies

- The dataset is imbalanced: 84% non-delinquent and 16% delinquent customers.
- Numerical features show weak correlations with delinquency.
- Payment history variables appear more informative than individual financial variables.
- Unemployed customers show relatively higher delinquency counts.

5. High-Risk Indicators

- Repeated late payments are strong indicators of future delinquency.
- Missed payments suggest repayment difficulties.
- Higher credit utilization may indicate financial stress.
- Higher debt-to-income ratios may increase default risk.
- Employment instability may negatively impact repayment capacity.

6. Conclusion

- The dataset is generally clean and suitable for predictive modeling after preprocessing.
- Historical payment behavior emerged as the strongest predictor of delinquency.
- Future modeling should address class imbalance using methods such as SMOTE or class weighting.